AML CENTRALIZED WORKFLOW MANAGEMENT SOLUTION

Business Requirements, Functional Specification, Data Model, Workflow Controls and AI Development Blueprint

Purpose of this document
A build-ready specification for a deterministic, auditable workflow platform that imports Predator alert extracts, centralizes officer and HOD activities, manages RFV/RFI, freeze and STR actions, and prevents information loss across the complete case lifecycle.

Document owner	Jabatan Pematuhan AML/CFT
Business sponsor	Head of Department
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Classification	Internal - Confidential
System type	Workflow and centralized case information platform
Implementation principle	Rule-based, human-decided, audit-defensible

Prepared as a system-development blueprint

Document Control

Version	Date	Status	Description
1.0	06/08/2026	Baseline blueprint	Initial complete specification based on confirmed operating process and assessment of the three workbooks.

Approval and Sign-off

Role	Name	Decision	Date
Business owner	To be assigned	Pending	
Head of Department	To be assigned	Pending	
Technology owner	To be assigned	Pending	
Information Security	To be assigned	Pending	
Data/Records owner	To be assigned	Pending	

Authoritative Inputs

Input	Confirmed purpose	System treatment
Predator alert extract	Alert population and workload allocation, including the PIC for each alert.	Read-only import; Predator remains authoritative for alert data and official disposition.
RFV & RFI Repository Master Listing 2026.xlsx	Follow-up, branch/customer information request, evidence receipt and assessment.	Migrated into controlled RFV/RFI module; export maintained if still required.
AMCP STR INFORMATION TABLE 2026.xlsx	Suspicious case, offence, STR preparation/submission and historical repository.	Migrated into controlled STR module; export maintained if still required.

Interpretation rule
Statements labelled CONFIRMED are directly established by the current process. Statements labelled CONFIGURABLE require business approval before production. The implementation must not silently convert configurable assumptions into mandatory rules.

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1. Executive Summary

The proposed solution is a centralized AML workflow management platform that complements, but does not replace, Predator. It imports the alert extract prepared by Officer 1, presents assigned work to the relevant AML Officers, records investigation routing, sends suspicious cases to the HOD review queue, creates and monitors directed actions, centralizes RFV/RFI and STR information, and prevents case closure while required information or actions remain outstanding.

Governing design principle
One centralized case, one complete history, one accountable owner at every stage, and no unrecorded handoff.

1.1 Business Outcomes

- Reduce repeated manual entry across the Alert, RFV/RFI and STR workbooks.
- Preserve Predator as the source of truth for alerts and official alert disposition.
- Give each officer a controlled queue of cases under their purview.
- Separate the officer's suspicious assessment and recommendation from the HOD's review and direction.
- Automatically route approved actions into RFV, RFI, freeze, further-assessment or STR workflows.
- Ensure all communications, documents, decisions, references and deadlines are linked to the same case.
- Prevent closure while mandatory data, approvals or directed actions remain incomplete.
- Provide management with real-time workload, ageing, status and completeness visibility.
- Preserve a tamper-evident history of every import, assignment, decision, change and export.

1.2 Solution Boundary

Inside the solution	Outside / remains authoritative elsewhere
Alert import, queue allocation, cross-reference, internal workflow, HOD direction, RFV/RFI tracking, freeze-action tracking, STR workflow tracking, documents, communications, dashboards, audit trail.	Predator alert generation, transaction investigation capabilities within Predator, actual account-freezing execution, regulatory STR submission platform unless a formally approved integration is later implemented.

2. Confirmed Current-State Process

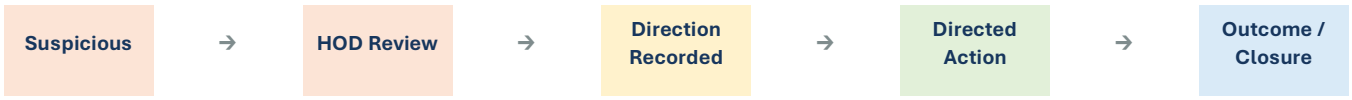
1. Officer 1 extracts the list of alerts from Predator, the Transaction Monitoring System.
2. Officer 1 shares the alert list with the AML team together with the assigned PIC for each alert.
3. Each assigned AML Officer investigates the alerts under their purview and determines whether each alert is suspicious or a false positive/non-suspicious case.
4. Cases categorized as suspicious by the AML Officer are referred to the Head of Department.
5. The HOD reviews the suspicious cases and advises the next action, which may include further assessment through RFV, freezing action, STR progression, another controlled action, or no further action.
6. Downstream information is currently maintained across separate RFV/RFI and AMCP STR workbooks.

2.1 Current-State Visual Workflow



Officer decision	Route
False positive / Non-suspicious	Complete Predator disposition and internal workflow closure controls.

Officer decision	Route
Suspicious	Mandatory HOD review; officer cannot independently finalize the downstream action.



3. Solution Vision and Scope

3.1 Target Operating Model

The system shall operate as an internal workflow and centralized information layer. Predator supplies alert data. The new platform controls allocation visibility, handoffs, directed actions, downstream repositories and audit history. It shall never infer or decide whether activity is suspicious.

3.2 In-Scope Capabilities

- Predator alert-extract import and batch validation
- PIC-based officer queues
- Unique Case ID generation and related-alert consolidation
- Customer/account historical checks
- Officer investigation progress and routing outcome
- Suspicious-case HOD review and direction
- RFV and RFI request, follow-up, response and evidence assessment
- Freeze-action tracking and dependency on STR/follow-up obligations
- STR drafting, review, approval and submission-status tracking
- Document and communication registers
- Workload, ageing, completeness and management dashboards
- Role-based access and immutable audit history
- Controlled Excel export and reconciliation reports

3.3 Explicitly Out of Scope for Initial Release

- Automated suspicious/non-suspicious classification
- Generative AI decision-making or risk scoring
- Automatic update of Predator without an approved API
- Automatic account freezing
- Automatic submission of an STR to a regulator
- Automatic sending of external email in the first release
- Deletion or overwrite of historical source records

3.4 Success Measures

Metric	Target / acceptance principle
Traceability	100% of imported alerts have batch, source file, sheet and row references.
Ownership	100% of active items have an assigned owner.
Handoff integrity	No suspicious case bypasses HOD review.

Metric	Target / acceptance principle
Completeness	No case closes while a mandatory field or open required action remains.
Duplicate control	Duplicate Alert IDs are blocked; potential historical matches are surfaced.
Auditability	Every material change records before value, after value, actor, timestamp and reason.
Reconciliation	Every import row is classified as imported, linked, rejected duplicate or exception.

4. Roles, Governance and Access

Role	Core responsibilities	Key permissions
Officer 1 / Alert Coordinator	Extract alert list, import the file, validate batch, confirm PIC allocation and resolve import exceptions.	Import; view all batch records; correct mapping through controlled reassignment; cannot alter source alert values.
AML Officer	Investigate assigned alerts in Predator; record progress, outcome, summary and recommendation; perform directed actions.	View own queue; update assigned cases; refer suspicious case; maintain assigned RFV/RFI, freeze or STR tasks.
Head of Department	Review suspicious cases and direct next action.	View all relevant case information; return, approve or direct action; set priority, owner and due date.
STR Reviewer/Approver	Review STR narrative and completeness.	Return for amendment; approve; record approval.
STR Submitter	Submit approved STR through approved reporting channel and record reference.	Update submission fields only after approval.
System Administrator	Maintain users, roles, reference data and configuration.	No authority to make AML case decisions unless separately assigned a business role.
Auditor / Read-only Reviewer	Inspect records and history.	Read-only access to permitted data and exportable audit evidence.

4.1 Segregation of Duties

- A system administrator must not gain business decision authority solely because of technical access.
- An officer's suspicious conclusion and the HOD's direction must be separate events with separate actors and timestamps.
- An STR cannot be marked Submitted unless required review/approval is recorded.
- Any reassignment must retain previous ownership history and a reason.
- Permission checks must occur server-side or in the trusted application layer, not only in the user interface.

5. End-to-End Workflow Specification

5.1 Master Workflow

ID	Stage	Entry condition	Owner	Exit
W01	Alert imported	Alert exists in validated import batch.	System/Officer 1	Assigned to PIC or import exception.
W02	Officer investigation	Officer reviews case in Predator and records	AML Officer	Non-suspicious or suspicious referral.

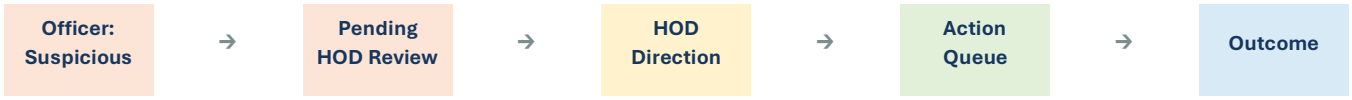
ID	Stage	Entry condition	Owner	Exit
		workflow status.		
W03	Suspicious referral	Mandatory findings and recommendation submitted.	AML Officer	Pending HOD review.
W04	HOD review	HOD reviews full context and records direction.	HOD	Directed action created or no-further-action route.
W05	Directed action	RFV/RFI, freeze, further assessment, STR, monitoring, link, return or other.	Assigned action PIC	Action outcome recorded.
W06	Return to HOD	Where suspicion or decision remains after action.	System/Officer	Final/new direction.
W07	Closure	All mandatory information and actions complete; official disposition confirmation recorded.	Authorized officer/HOD	Closed and locked except amendments.

5.2 Non-Suspicious Route



- The system shall require an assessment summary and investigation completion date.
- The system shall record whether the Predator disposition was completed and the date.
- If an exception, related action or document obligation is open, closure must be blocked.

5.3 Suspicious Route



- The officer may recommend an action but cannot record the HOD direction.
- HOD direction must include actor, date, decision, remarks, action PIC and due date where an action is required.
- If action outcome requires reconsideration, the case returns to HOD review rather than being routed directly by the officer.

5.4 HOD Direction Catalogue

Direction code	Meaning	System result
NFA	No further action	Create closure task and require final rationale/disposition confirmation.
RETURN	Return to officer	Create revision task with mandatory HOD remarks.
ASSESS	Further internal assessment	Create assessment task; return to HOD on completion.
RFV	Request for Verification	Create linked RFV record.
RFI	Request for Information	Create linked RFI record.
FREEZE	Initiate freeze action	Create linked freeze-action record; track actual execution externally.

Direction code	Meaning	System result
STR	Proceed with STR	Create linked STR case and drafting task.
LINK	Link to existing case	Require target Case ID and relationship reason.
MONITOR	Continue monitoring	Create time-bound monitoring action with review date.
OTHER	Other	Require description, owner, due date and approval note.

5.5 RFV/RFI Lifecycle

Draft → Approved / Authorized → Sent → Awaiting Response → Follow-up(s) → Response Received → Assessment → Completed

After assessment, the result shall either resolve the concern and enter closure review, or remain suspicious and return to the HOD review queue. The implementation shall not assume that RFV/RFI completion automatically means non-suspicious.

5.6 Freeze Lifecycle

Directed → Pending Initiation → Initiated → Pending Confirmation → Effective / Unsuccessful → Follow-up Obligations → Completed

5.7 STR Lifecycle

Pending Draft → Drafting → Draft Completed → Review → Amend / Approve → Pending Submission → Submitted → Acknowledged

6. Functional Requirements

Requirement ID	Module	Requirement	Priority
FR-IMP-001	Import	The system shall import an authorized .xlsx Predator extract without modifying the source file.	Must
FR-IMP-002	Import	The system shall record batch ID, filename, hash, sheet, row, importer and timestamp.	Must
FR-IMP-003	Import	The system shall validate required columns and reject or quarantine structurally invalid rows.	Must
FR-IMP-004	Import	The system shall block duplicate Alert IDs and report them as reconciled exceptions.	Must
FR-IMP-005	Import	The system shall preserve identifiers as text, including leading zeroes.	Must
FR-IMP-006	Import	The system shall preserve original raw values alongside normalized matching values.	Must
FR-CASE-001	Case	The system shall generate a unique immutable Case ID.	Must
FR-CASE-002	Case	The system shall permit multiple related Alert IDs to link to one Case ID.	Must

Requirement ID	Module	Requirement	Priority
FR-CASE-003	Case	The system shall display historical RFV/RFI and STR matches by customer ID and account number.	Must
FR-CASE-004	Case	Potential matches shall require human confirmation; the system shall not auto-merge based on name similarity alone.	Must
FR-CASE-005	Case	The system shall record the reason and actor for every case merge, link or unlink.	Must
FR-QUE-001	Queues	Each officer shall see alerts assigned to them by default.	Must
FR-QUE-002	Queues	HOD shall see suspicious cases pending review.	Must
FR-QUE-003	Queues	The system shall provide separate queues for RFV/RFI, freeze, STR, returned and overdue actions.	Must
FR-QUE-004	Queues	Supervisors shall be able to view workload by officer and stage.	Must
FR-INV-001	Investigation	The system shall record investigation start/completion dates, Predator reference, summary, outcome and recommendation.	Must
FR-INV-002	Investigation	Suspicious referral shall require grounds, suspected offence and completion date.	Must
FR-INV-003	Investigation	The system shall not make or recommend an AML suspicious decision using AI.	Must
FR-INV-004	Investigation	A non-suspicious route shall require Predator disposition confirmation before closure.	Must
FR-HOD-001	HOD	Every officer-suspicious case shall enter HOD review.	Must
FR-HOD-002	HOD	HOD direction shall be stored separately from officer recommendation.	Must
FR-HOD-003	HOD	Actions requiring execution shall require an action PIC and due date.	Must
FR-HOD-004	HOD	HOD shall be able to return a case for revision with mandatory remarks.	Must
FR-RFV-001	RFV/RFI	The system shall create an RFV/RFI record from the case without re-keying shared customer and alert fields.	Must
FR-RFV-002	RFV/RFI	The system shall track request, due, follow-up, response and assessment dates.	Must
FR-RFV-003	RFV/RFI	The system shall calculate overdue follow-up tasks using configured SLA rules.	Must
FR-RFV-004	RFV/RFI	Completion shall require response status, evidence assessment, final conclusion and next route.	Must
FR-FRZ-001	Freeze	The system shall track freeze direction, initiation, confirmation, reference, affected account and outcome.	Must
FR-FRZ-002	Freeze	The system shall not execute an account freeze.	Must
FR-FRZ-003	Freeze	The system shall track whether freeze activity creates an STR or other obligation.	Must
FR-STR-001	STR	The system shall generate an STR case from an approved direction and inherit linked context.	Must
FR-STR-002	STR	The system shall keep offence, reporting type, duplicate status, referral source and submission status as separate fields.	Must
FR-STR-003	STR	Submitted status shall require STR reference, submission date and submitter.	Must
FR-STR-004	STR	Submitted content shall be locked; corrections shall be recorded as amendments.	Must
FR-DOC-001	Documents	The system shall maintain a document register linked to Case ID and action.	Must

Requirement ID	Module	Requirement	Priority
FR-COM-001	Communications	The system shall maintain dated communication records and attachment references.	Must
FR-AUD-001	Audit	The system shall record before value, after value, actor, timestamp and reason for material changes.	Must
FR-CLS-001	Closure	The system shall block closure while required fields, actions, approvals or dependencies remain open.	Must
FR-EXP-001	Export	The system shall produce controlled RFV/RFI and STR exports if the existing Excel formats remain required.	Must
FR-REP-001	Reporting	The system shall provide operational, HOD and management dashboards with drill-through to case detail.	Must

7. Data Model and Data Dictionary

7.1 Logical Entity Model

Entity	Cardinality / relationship	Purpose
ImportBatch	1 to many AlertImportRows	Evidence of each Predator extract import.
Alert	Many alerts may link to one Case	Preserves source alert identity and allocation.
Case	Central parent entity	Connects all stages and history.
Customer	One customer may have many cases	Normalized customer identity; raw source retained.
Account	Many accounts may link to a case	Tracks affected accounts and roles.
Investigation	One or more per case over time	Officer workflow record and conclusion.
HODReview	One or more per suspicious cycle	Direction and governance event.
Action	Parent for directed tasks	Common ownership, priority, due date and status.
RFVRFI	Action subtype	Request, follow-up, response and assessment.
FreezeAction	Action subtype	Freeze tracking without execution.
STRCase	Action subtype	Drafting, review, approval and submission.
Communication	Many per case/action	Email/call/internal communication metadata.
Document	Many per case/action	Document inventory and verification.
AuditEvent	Many per record	Immutable change log.
ReferenceData	Configurable controlled values	Statuses, offences, users, SLA, branches.

7.2 Core Field Dictionary

Entity	Field	Type	Required	Definition / rule

Entity	Field	Type	Required	Definition / rule
Case	case_id	String	Yes	Immutable generated value; AML-YYYY-NNNNNN.
Case	current_stage	Enum	Yes	Derived from active workflow state.
Case	current_status	Enum	Yes	Controlled status; never free text.
Case	current_owner_user_id	FK	Yes for active	One accountable owner.
Case	final_outcome	Enum	At closure	Non-suspicious, STR submitted, linked, other approved outcome.
Alert	alert_id	String	Yes	Original Predator Alert ID; unique.
Alert	source_batch_id	FK	Yes	Import traceability.
Alert	source_row_no	Integer	Yes	Original row number.
Alert	customer_id_raw	String	Yes	Exact source value.
Alert	customer_id_normalized	String	Yes	Trimmed matching value; never replaces raw.
Alert	assigned_pic_raw	String	Yes	Source assignment value.
Alert	assigned_user_id	FK	Yes	Mapped internal user.
Investigation	officer_outcome	Enum	Yes at completion	NON_SUSPICIOUS or SUSPICIOUS.
Investigation	recommendation	Enum	For suspicious	Officer recommendation; not HOD decision.
Investigation	assessment_summary	Text	Yes	Concise record of review conclusion.
Investigation	grounds_for_suspicion	Text	For suspicious	Specific factual grounds.
Investigation	predator_disposition_confirmed	Boolean	At closure	Confirms official TMS action.
HODReview	direction_code	Enum	Yes	NFA, RETURN, ASSESS, RFV, RFI, FREEZE, STR, LINK, MONITOR, OTHER.
HODReview	direction_remarks	Text	Conditional	Mandatory for RETURN, OTHER and configured decisions.
HODReview	action_owner_id	FK	For action	Assigned executor.
HODReview	action_due_at	Datetime	For action	SLA due date.
RFVRFI	request_type	Enum	Yes	RFV or RFI.
RFVRFI	requested_information	Text	Yes	Specific request.
RFVRFI	sent_at	Datetime	At sent	Actual send timestamp.
RFVRFI	response_received_at	Datetime	Conditional	Actual receipt timestamp.
RFVRFI	evidence_assessment	Text	At completion	Officer evaluation.
RFVRFI	post_assessment_route	Enum	At completion	Resolved/closure or return to HOD.
FreezeAction	affected_account_id	FK	Yes	Account affected.
FreezeAction	external_reference	String	Conditional	Reference from approved process.

Entity	Field	Type	Required	Definition / rule
FreezeAction	effective_at	Datetime	If effective	Effective date/time.
FreezeAction	outcome	Enum	At decision	Effective, unsuccessful, rejected, lifted, etc.
STRCase	primary_offence_code	FK	Yes	Approved offence taxonomy.
STRCase	reporting_type	Enum	Yes	Initial, supplemental, defensive, other approved type.
STRCase	duplicate_status	Enum	Yes	Separate from submission status.
STRCase	grounds_for_suspicion	Text	Yes	Full STR rationale.
STRCase	str_reference	String	At submitted	Official reference.
STRCase	submitted_at	Datetime	At submitted	Submission timestamp.
Document	storage_uri	String	Yes	Controlled location/reference.
Document	verification_status	Enum	Yes	Not reviewed, verified, rejected, not applicable.
Communication	communication_type	Enum	Yes	Initial request, follow-up, response, HOD, etc.
AuditEvent	before_json	JSON	Conditional	Prior state.
AuditEvent	after_json	JSON	Yes	New state.
AuditEvent	reason	Text	For corrections	Mandatory for manual correction/reassignment.

7.3 Identifier Rules

- All customer IDs and account numbers shall be stored as strings, never numeric types.
- Raw source values and normalized matching values shall both be retained.
- Case ID, action IDs and internal record IDs are immutable.
- Alert ID is unique within the system. A duplicate import shall not create a second alert record.
- Name matching may support search but shall not be the sole basis of an automatic case link.

8. State Machines and Business Rules

8.1 Case States

State	Allowed next states
IMPORTED	ASSIGNED, IMPORT_EXCEPTION
ASSIGNED	UNDER_INVESTIGATION, REASSIGNED
UNDER_INVESTIGATION	NON_SUSPICIOUS_REVIEW, PENDING_HOD_REVIEW
PENDING_HOD_REVIEW	RETURNED_TO_OFFICER, ACTION_OPEN, CLOSURE_REVIEW
ACTION_OPEN	PENDING_HOD_REVIEW, CLOSURE_REVIEW, ACTION_OPEN
CLOSURE_REVIEW	CLOSED, ACTION_OPEN, RETURNED_TO_OFFICER

State	Allowed next states
CLOSED	AMENDMENT_OPEN only through authorized process
IMPORT_EXCEPTION	IMPORTED after correction or REJECTED

8.2 Critical Business Rules

Rule ID	Rule
BR-001	Predator is authoritative for alert content and official alert disposition.
BR-002	The source alert extract must be immutable after import.
BR-003	A suspicious officer outcome always requires HOD review.
BR-004	Officer recommendation does not equal HOD direction.
BR-005	The system must not auto-classify suspiciousness or generate a decision.
BR-006	A directed action must have one accountable owner.
BR-007	An action cannot be completed without its subtype-specific mandatory fields.
BR-008	RFV/RFI completion does not automatically resolve suspicion.
BR-009	Freeze tracking does not constitute execution of the freeze.
BR-010	STR status, offence, reporting type and duplicate status are independent fields.
BR-011	A submitted STR record is immutable; amendments are additive.
BR-012	Case closure is blocked while any required action or field remains open.
BR-013	Every manual correction or reassignment requires a reason and is audited.
BR-014	Potential historical matches require human confirmation.
BR-015	All date/time values must use a consistent timezone and preserve original source dates.
BR-016	Export totals must reconcile to the centralized repository as of the export timestamp.

9. Validation and Completeness Controls

9.1 Stage Gates

Gate	Mandatory conditions
Suspicious referral	Assessment summary; grounds for suspicion; suspected offence; investigation completion date; officer recommendation or explicit no recommendation.
HOD direction	Reviewer; review date; direction; remarks where required; action owner and due date where applicable.

Gate	Mandatory conditions
RFV/RFI sent	Request type; branch/recipient; requested information; sent date; PIC; response due date.
RFV/RFI completed	Response status; evidence assessment; final conclusion; post-assessment route.
Freeze effective/completed	Affected account; action reference or recorded reason unavailable; status; dates; outcome; related obligations reviewed.
STR submitted	Required approvals; primary offence; grounds; review period; STR PIC; official reference; submission date; submitter.
Case closed	No open required action; HOD direction satisfied; Predator disposition confirmation; final outcome; closure rationale; closing actor/date.

9.2 Completeness Engine

The system shall show both a diagnostic completeness percentage and a binary control result. The binary result governs workflow progression.

Applicable Field Set → **Mandatory Checks** → **Dependency Checks** → **Outstanding Items** → **COMPLETE / INCOMPLETE**

No silent defaults
The system must not fill missing operational decisions with assumptions such as Pending, No or Not Applicable. Defaults may be used only for system-generated metadata and approved reference values.

10. User Experience Specification

10.1 Navigation

- My Work
- All Cases (authorized roles)
- Import Batches
- HOD Review
- RFV/RFI
- Freeze Actions
- STR Cases
- Documents and Communications
- Dashboards
- Reports and Exports
- Administration

10.2 My Work Queue Columns

Column	Behaviour
Case ID	Clickable link.
Alert ID(s)	Shows count and expandable list.

Column	Behaviour
Customer	Name plus masked/authorized identifier.
Stage / Status	Controlled badges.
Owner	Current accountable user.
Priority	Configured visual indicator.
Due / Ageing	Shows due date and overdue status.
Historical Match	No match, open RFV/RFI, prior STR, multiple matches.
Completeness	Complete or outstanding-item count.
Next Action	Single clear required action.

10.3 Case Workspace Tabs

- Overview
- Customer and Accounts
- Predator Alerts
- Investigation
- HOD Reviews
- Actions
- RFV/RFI
- Freeze
- STR
- Documents
- Communications
- History

10.4 Usability Controls

- Save draft without allowing stage transition when mandatory information is incomplete.
- Show validation messages next to the field and in a consolidated error summary.
- Require confirmation only for high-impact actions such as referral, HOD direction, merge/link, submission and closure.
- Use search by Case ID, Alert ID, customer ID, account number, STR reference and RFV/RFI ID.
- Never rely on color alone to communicate status.

11. Reports, Dashboards and Notifications

11.1 Dashboard Measures

Audience	Required measures
Officer	New assigned alerts; in investigation; returned; RFV/RFI due; responses received; freeze tasks; STR drafts; overdue; incomplete records.
HOD	Suspicious cases pending review; directions by type; high priority; prior STR matches; RFV/RFI outcomes; STR approvals; overdue actions.

Audience	Required measures
Management	Alerts imported; workload by PIC; outcomes; HOD direction breakdown; RFV/RFI ageing; freeze actions; STR progress; processing time; SLA breaches; completeness exceptions.
Control/Audit	Import reconciliation; data changes; reassignments; merges/links; closure overrides attempted; export reconciliation; access events.

11.2 Notification Events

- New alert assigned
- Suspicious case submitted to HOD
- Case returned for revision
- Directed action assigned
- Action due soon / overdue
- RFV/RFI follow-up due
- Response received but not assessed
- Freeze pending confirmation
- STR returned for amendment
- Approved STR pending submission
- Case closure blocked by outstanding items

Initial release should use in-application notifications and optional Outlook draft generation. Automatic external email sending is out of scope until separately approved.

12. Integration, Import and Export

12.1 Predator Import Contract

Aspect	Requirement
Format	.xlsx; worksheet and header mapping configurable.
Required fields	Alert ID, customer ID, customer name, assigned PIC; other fields validated per approved mapping.
Idempotency	Same Alert ID cannot create more than one alert.
Lineage	Batch, filename, hash, worksheet and row number retained.
Normalization	Whitespace and non-breaking spaces handled; leading zeroes preserved; raw values retained.
Failure	Invalid rows quarantined with reason; successful rows continue if batch policy permits.
Reconciliation	Total source data rows = imported + duplicate + exception + intentionally ignored summary rows.

12.2 Existing Workbook Treatment

During migration and parallel testing, the RFV/RFI and AMCP workbooks shall remain controlled inputs. After acceptance, the centralized repository becomes the operational source for downstream workflow information, and Excel files may be generated as controlled outputs where the existing format is still required.

12.3 Export Requirements

- Export timestamp and as-of date
- Exported by
- Source query/filter
- Row count
- Control total or hash where applicable
- Template version
- Successful/failed status
- Link to export artifact
- Reconciliation to source records

13. Security, Audit and Records Controls

13.1 Security Principles

- Least privilege and role-based access
- Named user accounts only
- Encryption in transit and at rest where supported
- Secure secrets management; no credentials in code or configuration files
- Protection of personal and financial information in screen, export and logs
- Session timeout and failed-login controls
- Environment segregation for development, testing and production
- Backup and tested restoration procedures

13.2 Audit Events

Event category	Examples
Authentication	Login, logout, failed access, role change.
Import	Batch created, file hash, rows imported/rejected.
Case	Created, linked, merged, unlinked, reassigned, state changed.
Decision	Officer outcome, recommendation, HOD direction, approval.
Action	Created, assigned, due changed, completed, reopened.
Data	Before/after field values and reason.
Document	Added, replaced, verified, rejected, accessed where required.
Export	Report name, filters, row count, actor, timestamp.
Administration	Reference data, SLA, user and permission changes.

13.3 Retention

Retention periods, legal holds, archival rules and deletion authority are CONFIGURABLE and must be approved by the Records, Legal, Compliance and Information Security owners. The build shall support retention policies but shall not assume a period.

14. Non-Functional Requirements

ID	Quality	Requirement
NFR-001	Availability	Business-hours availability target is configurable; planned downtime shall be communicated.
NFR-002	Performance	Standard queue and case screens should load within 3 seconds under agreed normal load.
NFR-003	Import	A typical alert batch shall import with validation and reconciliation within an agreed operational threshold.
NFR-004	Scalability	Design shall support multi-year history without requiring monthly database separation.
NFR-005	Reliability	Transactions shall be atomic; partial workflow transitions are prohibited.
NFR-006	Recoverability	Automated backups and documented restore tests are mandatory.
NFR-007	Auditability	Audit history must be queryable and exportable without altering source records.
NFR-008	Maintainability	Status, offence, SLA, branch and user mappings shall be configuration-driven.
NFR-009	Accessibility	Keyboard navigation, readable contrast, labels and non-color status indicators.
NFR-010	Localization	Dates displayed as dd/mm/yyyy; timestamps stored consistently with timezone.
NFR-011	Privacy	Sensitive fields masked according to role and export policy.
NFR-012	Determinism	Workflow decisions and validation shall be rule-based and reproducible.

15. Migration and Reconciliation

15.1 Migration Sequence

7. Freeze source copies and calculate file hashes.
8. Profile each workbook and identify operational rows, summaries, duplicates and malformed values.
9. Define field mapping and controlled-value mapping.
10. Load raw staging records with file/sheet/row lineage.
11. Normalize identifiers without overwriting raw values.
12. Load customers, accounts, cases, RFV/RFI and STR records.
13. Run exact and potential duplicate rules.
14. Produce unresolved-match worklist for business review.
15. Reconcile row counts and key totals.
16. Obtain business sign-off before production cutover.

15.2 Known Data Issues to Handle

- Mixed date representations, including Excel serial numbers and text dates.
- Customer IDs and account numbers at risk of losing leading zeroes.
- Summary/pivot rows embedded in the RFV/RFI operational worksheet.
- Repeated customers and accounts that may represent legitimate repeat reviews, additional accounts or duplicates.
- Fragmented STR status and offence descriptions.
- Multiple monthly/repository worksheets with potentially overlapping records.
- Blank workflow fields that must not be interpreted automatically.

15.3 Reconciliation Formulae

Import control
Source operational rows = Imported new records + Linked existing records + Confirmed duplicate records + Quarantined exceptions + Approved ignored non-operational rows.

Suspicious-case control
Officer suspicious referrals = Pending HOD review + HOD returned + Open directed actions + Closure reviews + Closed cases.

STR control
HOD STR directions = STR cases created + Linked existing STRs + Approved not-proceeded outcomes, with every difference individually explained.

16. Testing and Acceptance

16.1 Test Layers

- Unit tests for normalization, validation, state transitions and permissions.
- Integration tests for workbook import/export, database transactions and notifications.
- Migration tests for row lineage, duplicate handling and counts.
- Security tests for role enforcement and sensitive-data exposure.
- Performance tests using representative and peak volumes.
- Recovery tests for backup restoration and interrupted import.
- User acceptance testing for Officer 1, AML Officer, HOD, STR roles and audit view.

16.2 Critical Acceptance Scenarios

ID	Scenario	Expected result
AT-001	Import a valid Predator file	All rows reconciled; source unchanged; queues created.
AT-002	Re-import same file	No duplicate alert; batch reports duplicates.
AT-003	Identifier with leading zero	Value preserved exactly and searchable.
AT-004	Officer marks suspicious	Case enters HOD queue; downstream route blocked until HOD decision.
AT-005	Officer recommends RFV; HOD directs freeze	Both recommendation and differing direction preserved.
AT-006	RFV response incomplete	Completion blocked with specific missing-field messages.
AT-007	RFV concern remains	Case returns to HOD review.
AT-008	Freeze effective but STR obligation open	Case closure blocked.
AT-009	STR marked submitted without reference	Transition blocked.
AT-010	Attempt to edit submitted STR	Edit blocked; amendment route offered.
AT-011	Potential customer history match	Human confirmation required; no auto-merge.

ID	Scenario	Expected result
AT-012	Case closure with open action	Closure blocked and open items listed.
AT-013	Unauthorized user opens HOD decision	Access denied and event logged.
AT-014	Export RFV/RFI register	Rows and control totals reconcile to repository.
AT-015	Audit reconstruction	Reviewer can reconstruct all decisions and handoffs chronologically.

16.3 Definition of Done

- All Must requirements implemented and tested.
- No unresolved critical or high-severity defects.
- Migration and reconciliation signed off.
- Role and segregation controls accepted.
- UAT completed by every operational role.
- Backup and restore test passed.
- Operational handbook and support model delivered.
- Production deployment and rollback plan approved.

17. Deployment and Operating Model

17.1 Recommended Architecture

Layer	Initial recommendation	Future-ready option
Client	Internal desktop or intranet web interface.	Enterprise web application.
Application	Modular rule-based service layer.	API-based services.
Database	SQLite only for controlled single-user/pilot; otherwise managed relational database.	SQL Server or approved enterprise database.
Files	Controlled document store with references in database.	SharePoint/approved records repository.
Authentication	Approved organizational identity where available.	Microsoft Entra ID with role groups.
Email	Outlook draft generation only initially.	Approved Graph/API workflow.

Concurrency decision
If multiple officers must update the system simultaneously, a shared SQLite file is not the recommended production design. Use an approved multi-user relational database and application service.

17.2 Environments

- Development: synthetic or anonymized data only.
- Test/UAT: controlled representative data with access restricted.
- Production: approved infrastructure, backup, monitoring and change control.

17.3 Operational Controls

- Daily import monitoring
- Failed-job and exception review
- User access review
- Reference-data change approval
- Backup monitoring and periodic restore test
- Data-quality dashboard review
- Monthly reconciliation and audit-log archive
- Documented incident and rollback procedure

18. AI Developer Handoff Contract

Instruction to any implementing AI
Implement only the confirmed requirements and explicitly approved configuration. Do not infer regulatory obligations, authority levels, SLA periods, retention periods, offence taxonomies or external integrations that are not specified. Where information is missing, expose it as a configuration or raise a documented clarification item.

18.1 Required Deliverables from the Implementer

17. Solution architecture and threat model
18. Database schema and migration scripts
19. Application source code with modular services
20. Configuration files and reference-data seed files
21. Automated tests mapped to requirement IDs
22. Import/export mapping and reconciliation reports
23. Role-permission matrix implementation
24. Audit-event specification and sample reconstruction
25. Deployment, backup, restore and rollback scripts
26. Administrator, Officer, HOD, STR and Support manuals
27. UAT pack with evidence
28. Known limitations and open decisions register

18.2 Implementation Constraints

- No AI-based suspicious decision or autonomous routing beyond approved deterministic workflow rules.
- No modification of the source Predator extract.
- No silent correction of IDs, dates, statuses or decisions.
- No hard-coded officer names, branches, offences, SLA or file paths.
- No destructive deletion of case history.
- No direct external action such as freeze, email send or STR submission without approved integration and authorization.
- All database transitions must be transactional and audited.

18.3 Suggested Module Boundaries

Module	Responsibility
identity_access	Authentication, roles, permissions and masking.
imports	File validation, staging, lineage, normalization and reconciliation.
case_core	Case, customer, account, alert links and ownership.

Module	Responsibility
investigation	Officer workflow and Predator disposition confirmation.
hod_review	Review queue, directions and returns.
actions	Generic task engine, due dates and dependencies.
rfv_rfi	Requests, follow-ups, responses and evidence assessment.
freeze	Freeze tracking and obligations.
str_workflow	Drafting, review, approval, submission and amendment.
documents	Document metadata and controlled references.
communications	Communication events and draft generation.
reporting	Dashboards, reports, exports and reconciliations.
audit	Immutable business and technical events.
configuration	Controlled lists, mappings, SLA, templates and features.

18.4 API / Service Behaviour Principles

- Commands change state; queries do not.
- Every state-changing command includes actor, expected current version and reason where required.
- Use optimistic concurrency or equivalent control to prevent lost updates.
- Return structured validation errors with requirement/rule identifiers.
- Idempotency keys are required for imports and external callback events.
- Do not expose raw sensitive data in logs or error messages.

18.5 Minimum Repository Structure

```
/src
  /identity_access
  /imports
  /case_core
  /investigation
  /hod_review
  /actions
  /rfv_rfi
  /freeze
  /str_workflow
  /documents
  /communications
  /reporting
  /audit
  /configuration
/tests
/migrations
/config
/docs
/deploy
/scripts
```

Appendix A. Controlled Value Catalogue

Domain	Initial controlled values
Officer outcome	NON_SUSPICIOUS; SUSPICIOUS; RETURNED_REASSIGNED; UNABLE_TO_COMPLETE
HOD direction	NFA; RETURN; ASSESS; RFV; RFI; FREEZE; STR; LINK; MONITOR; OTHER
RFV/RFI status	DRAFT; SENT; AWAITING_RESPONSE; FOLLOW_UP_1; FOLLOW_UP_2; RESPONSE_RECEIVED; UNDER_ASSESSMENT; COMPLETED; CLOSED
Freeze status	DIRECTED; PENDING_INITIATION; INITIATED; PENDING_CONFIRMATION; EFFECTIVE; UNSUCCESSFUL; REJECTED; LIFTED; COMPLETED
STR status	PENDING_DRAFT; DRAFTING; DRAFT_COMPLETED; PENDING_REVIEW; RETURNED_AMENDMENT; APPROVED; PENDING_SUBMISSION; SUBMITTED; ACKNOWLEDGED; NOT_PROCEEDED; WITHDRAWN
Duplicate status	NOT_DUPLICATE; POTENTIAL_DUPLICATE; CONFIRMED_DUPLICATE; LINKED_EXISTING_STR
Completeness	COMPLETE; INCOMPLETE

Appendix B. Configuration Decisions Required Before Build Completion

Decision	Owner	Required outcome
Exact meaning and use of RFV versus RFI	Business owner	Definitions and selection rules.
HOD direction list and authority limits	HOD/Compliance	Approved list and any additional approvals.
Freeze workflow and references	Operations/Legal/Compliance	Exact steps, evidence and dependencies.
STR review and approval hierarchy	Compliance	Roles and segregation rules.
Supported offence taxonomy	Compliance	Controlled codes and descriptions.
SLA and reminder periods	Business owner	Due-date rules by task.
Retention and archival periods	Records/Legal	Policy by record type.
Production technology and hosting	IT/Security	Approved architecture.
Authentication method	IT/Security	Identity and group model.
Document storage location	Records/IT	Approved repository and path convention.
Whether Excel exports remain official	Business owner	Transition and cutover decision.
Whether Outlook drafts are required	Business owner/IT	Template and access decision.

Appendix C. Visual Information Flow



HOD direction	Linked module	Returns to
No further action	Closure review	Closed when complete
Further assessment	Action task	HOD review
RFV/RFI	RFV/RFI module	HOD review or closure
Freeze	Freeze module	Open obligations / HOD / closure
STR	STR module	Submission / acknowledgement / closure
Link existing	Case relationship	HOD or closure
Monitoring	Time-bound task	HOD review

Appendix D. Mermaid Reference Diagram

The following machine-readable diagram may be used by developers or documentation tools. It is illustrative and must remain consistent with the requirements and state rules in this document.

```
flowchart TD
  P[Predator TMS] --> E[Officer 1 extracts alert list]
  E --> I[Import and validate batch]
  I --> Q[PIC-based officer queue]
  Q --> V[Officer investigates in Predator]
  V --> D{Officer outcome}
  D -->|Non-suspicious| C1[Confirm Predator disposition]
  C1 --> G[Completeness gate]
  D -->|Suspicious| H[HOD review]
  H --> A{HOD direction}
  A -->|Return| V
  A -->|RFV/RFI| R[RFV/RFI workflow]
  A -->|Freeze| F[Freeze action tracking]
  A -->|STR| S[STR workflow]
  A -->|Further assessment| X[Assessment task]
  A -->|No further action| G
  R --> O{Concern resolved?}
  O -->|No| H
  O -->|Yes| G
  F --> K[Track obligations]
  K --> H
  S --> T[Submitted / acknowledged]
  T --> G
  X --> H
  G --> Z{All mandatory data and actions complete?}
  Z -->|No| W[Return to responsible queue]
  Z -->|Yes| CL[Close and lock case]
```

Appendix E. Final Build Readiness Checklist

☐ Business definitions approved

☐ Role and authority matrix approved

☐ Status and offence catalogue approved

☐ Field mapping signed off

☐ SLA approved

☐ Security architecture approved

☐ Migration samples reconciled

☐ UI prototype approved

☐ Test cases accepted

☐ Backup and recovery demonstrated

☐ UAT signed off

☐ Production support model approved